



HS 508: Socio-technical Systems Engineering

Course Code	HS 508
Course Name	Socio-technical Systems Engineering
L-T-P-C	2-1-0-3
Prerequisites	none
Intended for	B. Tech./M.S./Ph.D.
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

28 Lectures to cover the main topics (approx 2 Lectures per week) 4 Tutorials/workshops TOTAL: 3 hrs per week, plus assessment time as required

- **Introduction to systems science and engineering**
 - Principles of systems science
 - Overview of systems engineering
 - Lifecycle, and other system views
- **Socio technical systems**
 - Background
 - Evolution and key issues
 - Overview of the state of the art
- **Knowledge Systems, the Web**
 - The role of knowledge in STS (Knowledge representation, modelling)
 - Distributed /Intelligent systems design
 - Collective Intelligence and Social networks
- **Task Analysis**
 - Traditional Task Decomposition
 - Knowledge/cognitive modelling of tasks
 - Cognitive walkthroughs
- **User centered design**
 - Purpose and approaches
 - Tools and techniques
- **Human computer interaction**
 - History of HCI
 - User Models
 - Interface Design
- **Interaction design**
 - Fundamentals of interaction design
 - Types of interactions
 - Handling dynamic complexity
- **STS Requirements**
 - Stakeholder analysis and management
 - Understanding social behaviors

Text Books

1. William Pasmore and John Sherwood. (editors), Principles of Sociotechnical Design, Sociotechnical Systems: A Sourcebook,
2. Somerville et al Whitworth, Brian with Ahmad, Adnan, Socio Technical Systems Engineering Handbook: SocioTechnical (available online), 2013.
3. Soegaard, Mads with Dam, Rikke Friis (eds.)System Design. In: The Encyclopedia of Human, Computer Interaction, 2nd Ed.. Aarhus, Denmark: The Interaction Design Foundation. Available online at
4. http://www.interactiondesign.org/encyclopedia/sociotechnical_system_design.html

References

1. Stevens Institute of Technology
2. The Knowledge Level Reinterpreted: Modeling Socio Technical Systems William J. Clancey